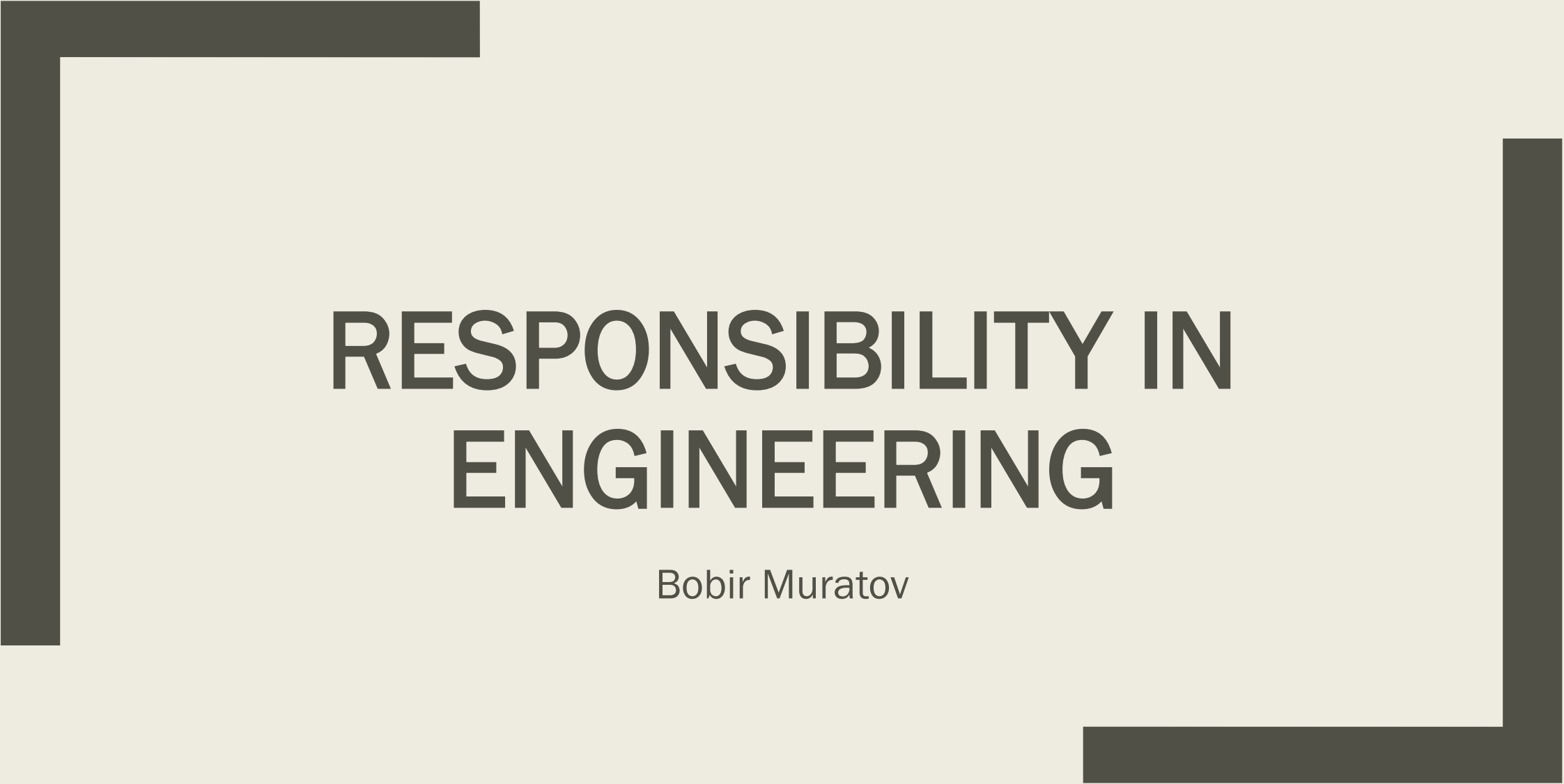
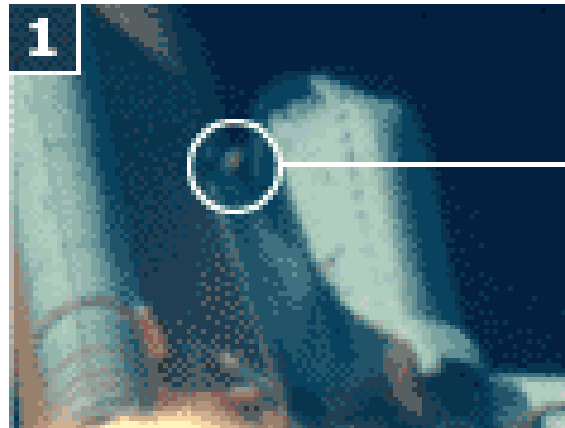


A decorative frame consisting of two thick, dark gray L-shaped lines. One L-shape is positioned on the left side, with its vertical bar extending downwards and its horizontal bar extending to the right. The other L-shape is on the right side, with its vertical bar extending upwards and its horizontal bar extending to the left. These two shapes meet at the center, framing the text.

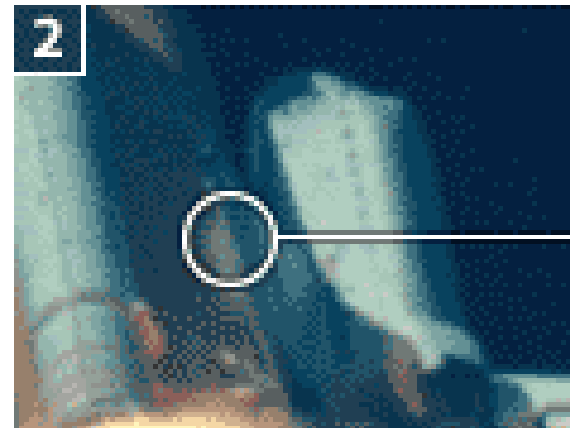
RESPONSIBILITY IN ENGINEERING

Bobir Muratov

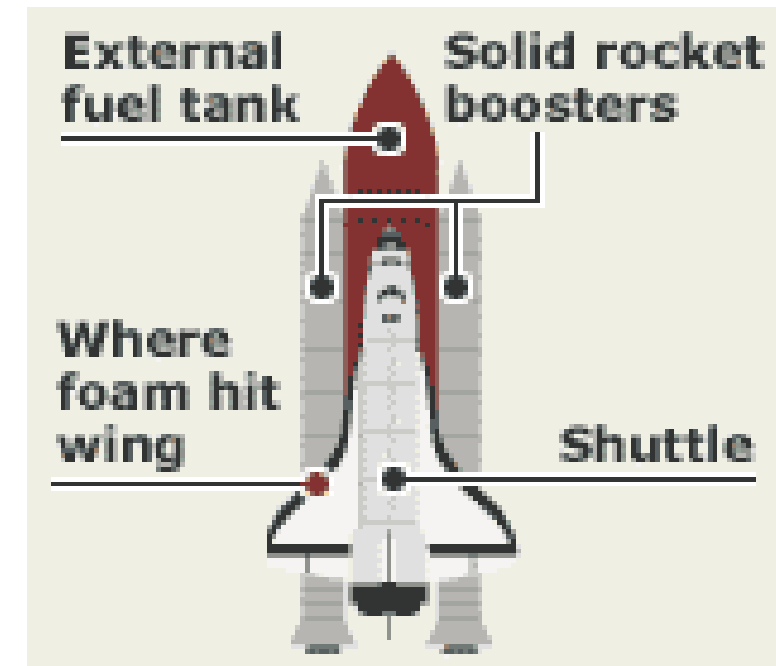
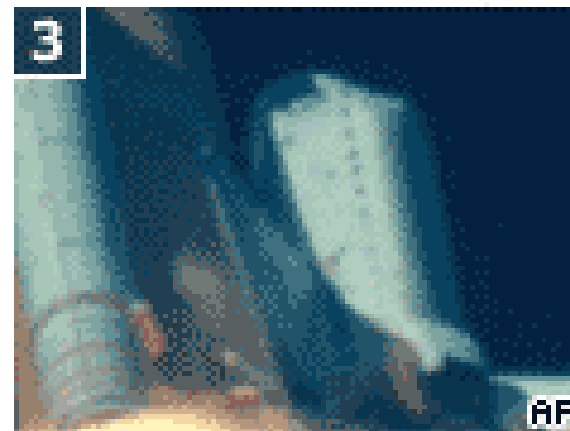
Columbia case



Piece of lightweight insulating foam breaks off fuel tank



Foam hits left wing and disintegrates



What is Responsibility?

Introduction

- “engineers shall conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession”

BOX 3.1 Responsibility as Accountability

Applied to:

- individual engineers;
- teams of engineers;
- divisions or units within organizations;
- organizations themselves.

Understood in terms of:

- legal accountability (which sometimes includes *strict* [no fault] liability);
- moral accountability (which does not include *strict* [no fault] liability).

Engineering Standards

- The work of engineers must satisfy applicable engineering standards
- obligation-responsibility
- blame-responsibility

BOX 3.3 Applicable Engineering Standards

- Regulatory: specifying technical requirements (e.g., for safety)
- Procedural: e.g., procedures to be followed for determining measurable quality or level of safety
- Standard of Care: that level or quality of service ordinarily provided by other normally competent practitioners, contemporaneously providing similar services in the same locality and under the same circumstances⁴
- Judgment: needed because regulatory and procedural standards, and the standard of care still require the exercise of good judgment

The Standard of Care

- Is it not enough to follow standard operating procedures and regulations?
- Do standards of safety and quality cover all aspects? Space for discretion?
- “An engineer is not liable, or responsible, for damages for every error. Society has decided, through case law, that when you hire an engineer, you buy the engineer’s normal errors”.
- Standard of care might best be seen as representing the highest shared standard among competent, responsible engineers in the relevant areas of practice

Blame Responsibility and Causation

- Columbia Accident Investigation Board identified two causes:
- the physical cause
- the organizational causes

Can Organizations also be Morally Responsible Agents?

Can Organizations also be Morally Responsible Agents?

- Philosopher Peter French:
- Corporations, like people, have a decision-making mechanism
- Corporations, like people, have policies that guide their decision-making
- Corporations, like people, have interests

BOX 3.5 Holding Organizations Responsible

1. For causing harms
2. For making reparations for wrong done
3. For making reforms

Harms: Legal Liability and Moral Responsibility

- Liability for harm implies that the person caused the harm, but it also implies something about the conditions under which the harm was caused

BOX 3.6 Legal and Moral Responsibility for Causing Harm

- Intentionally (or knowingly and deliberately) causing harm
- Recklessly causing harm—awareness of likelihood of causing harm, but not intending or aiming at harm
- Negligently causing harm—overlooking or not noticing risk of harm, failure of due care
- Strict liability for causing harm, even without fault: legal but not moral liability

What are the Impediments to
Responsibility?

Impediments to Responsibility

- The problem of many hands (or fractured responsibility)
- Blind spots
 - self-deception
 - willful blindness
 - inattentional blindness
- Normalizing deviance
- Egoistic perspectives (self-interest first)
- Egocentric perspectives (assuming others see matters as we do)
- Microscopic vision (seeing fine details, but missing the bigger picture)
- Uncritical deference to authority
- Groupthink
 - illusion of invulnerability of group
 - shared stereotypes
 - rationalizations
 - illusion of morality
 - self-censorship
 - illusion of unanimity
 - direct pressure to agree
 - mind-guarding (keeping dissenters away from the group)